Assignment 7: Fund X and the Two Regime Model

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What the model tells us

The two-regime idea suits Fund X because its returns change in both size and variability. Many months have modest gains, while other periods contain much larger gains. Letting the fund move between two return patterns describes that history better than forcing every month into one distribution.

The supplied note's exact fit is not the best one found. A fresh fit, independently checked with a second implementation, raises the log-likelihood from the reported 255.75 to **261.11**. It describes a quieter state averaging **1.55%** per month and a more variable state averaging **5.68%**. Both last about 13 months on average. That differs substantially from the note's short, two-month bursts averaging 9.6%.

Why that is economically plausible

The fund's factsheet describes a systematic long/short equity strategy with low net market exposure. Trading opportunities could change over time even when the overall market direction is unhelpful. Two positive-mean states are consistent with that description. Returns alone do not establish the trading mechanism, manager skill, or the accuracy of reported valuations. The model is useful for describing varying return conditions and building conditional scenarios.

What was checked and why

All 115 returns from January 2017 through July 2026 match the saved source factsheet exactly. There are five negative months; the mean is 3.52% and the sample standard deviation is 3.98% per month. July is marked estimated. The comparison keeps this date range fixed.

Table 1: Allowing two distributions produces most of the improvement

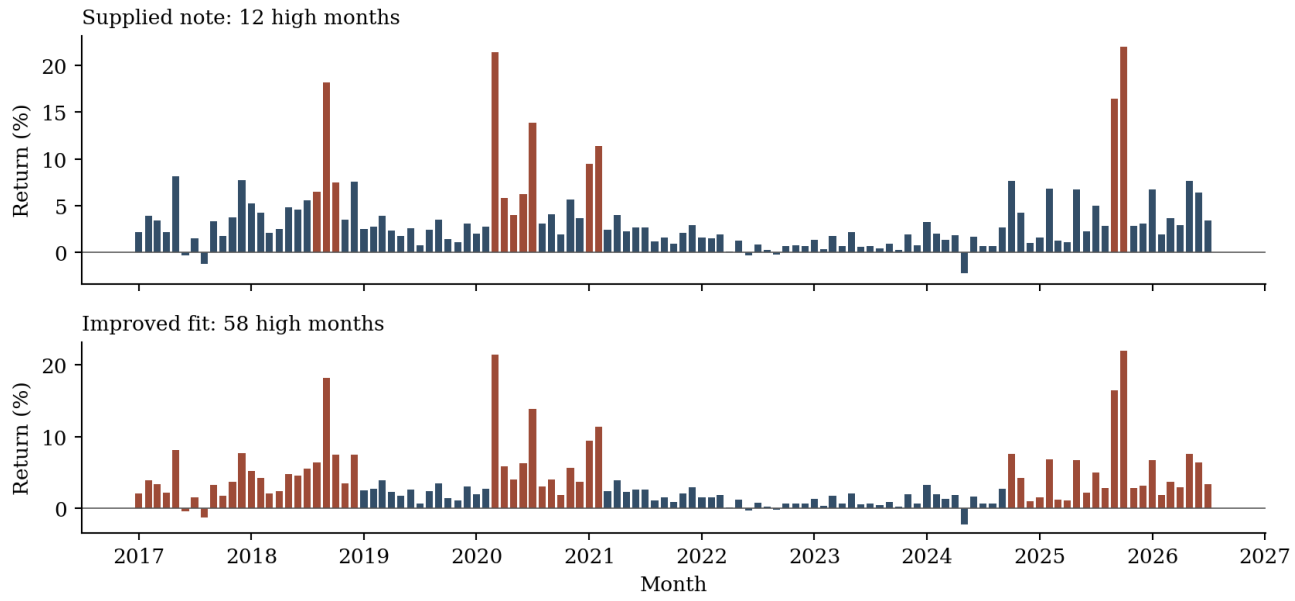
Model	Parameters	Log-likelihood	AIC	BIC
One regime	2	217.17	-430.34	-424.85
Independent mixture	5	256.91	-503.82	-490.10
Two regimes stationary	6	261.11	-510.22	-493.75
Two regimes free initial	7	261.39	-508.78	-489.56

An independent mixture allows two distributions but redraws the state each month. It already beats the note's reported likelihood. Adding persistence improves the refitted model further. Lower AIC and BIC favor the improved switching fit, but these are model comparisons, not proof of a unique economic explanation.

Table 2: The interpretation changes when the model is refitted

State	Mean return	Return SD	Duration (months)	Long-run share
Note ordinary	2.42%	2.16%	11.4	84.2%
Note high	9.56%	6.65%	2.1	15.8%
Improved lower	1.55%	1.16%	13.7	50.7%
Improved higher	5.68%	4.58%	13.3	49.3%

The high state is more than a handful of exceptional months



Red bars mark the higher-return state on the most likely full-sample path. These labels use hindsight. They are not signals available before each month. Long-run state shares in the table are different from observed path counts.

The tests and what they mean

Does the order of months matter? Yes. Shuffling the same 115 returns 499 times and refitting produces an equal or larger switching-versus-mixture gain only twice (one-sided Monte Carlo $p = 0.006$). Timing contains information beyond the return histogram. This rejects exchangeable ordering; gradual changes or other dependence could also produce that finding.

Does the model explain the rare losses? Only partly. The note's fitted distribution expects about **13.9** loss months out of 115; the improved fit expects **11.2**. Only **five** occurred. A positive mean in each state does not remove the negative tail. Neither model shows that future losses will remain rare or small.

Is dependence left over? Raw lag-one autocorrelation is 0.352. Predictive innovations from the improved fit have lag-one autocorrelation 0.156. Their 12-lag statistic is 19.70; a bootstrap that simulates and refits the model gives $p = 0.080$ (199 simulations). This does not reject at 5%, but it also does not establish independence or correct tail behavior.

Table 3: One-month-ahead checks from January 2025 through July 2026

Model	Total log predictive score	Mean forecast RMSE
One regime	24.15	5.66%
Independent mixture	32.99	5.66%
Switching model	26.90	5.49%

Higher log scores mean better predictions of the whole return distribution; lower RMSE means smaller errors in the predicted mean. The switching model has slightly lower mean error, but the simpler mixture has the best distribution score. Each forecast uses only earlier returns. This is a retrospective check of 19 months, with the model family chosen after seeing the full history, so it does not establish a reliable trading advantage.

Short technical notes

Model and units. Monthly simple returns are decimals, net of management and performance fees for the source's I series. With $y_t = \log(1 + r_t)$, $y_t \mid S_t = i \sim N(\mu_i, \sigma_i^2)$, and the two-state transition matrix is row-stochastic. The primary likelihood starts in its stationary distribution. The mean simple return is $e^{\mu_i + \sigma_i^2/2} - 1$; expected duration is $1/(1 - p_{ii})$. All reported likelihoods use log-return density. Converting to simple-return density subtracts $\sum_t \log(1 + r_t)$ from every likelihood, so model rankings and likelihood differences are unchanged. No annualization is used.

Optimization and reproduction. The primary search uses 300 starts, EM starting values, and direct constrained likelihood maximization. A separate `statsmodels` Hamilton-filter fit agrees within 10^{-5} log-likelihood units. The standard deviation floor is 0.001 in log-return units; halving and doubling it leaves the main result unchanged. The fitted standard deviations are far above this floor. A floor avoids a component collapsing onto one observation, where an unrestricted Gaussian-mixture likelihood can diverge. This is the best nondegenerate fit found, not a proof of a global maximum.

Corrections to the supplied note. Its one-regime likelihood, raw ACF, 12 high-month labels, and regime-demeaned lag-one ACF of 0.157 reproduce. Using its rounded parameters gives likelihood 255.56 with its stated rounded stationary weights, or 255.71 if the first month is fixed in the lower state. The original code, unrounded parameters, and initialization were not supplied, so the exact source of the reported 255.75 and stopping rule cannot be verified. In either case, better fits exist. Six parameters versus two means **four**, not three, extra parameters. Estimating the initial-state weight adds a seventh parameter. A usual chi-square likelihood-ratio reference is invalid when states become indistinguishable under the null. The note correctly recognizes nonregularity, but its one-regime comparison alone cannot isolate the contribution from persistence.

Residual and tail interpretation. Subtracting full-sample Viterbi state means uses hindsight and leaves different variances across states. Its ordinary Ljung–Box p-values are descriptive, without a valid automatic correction for estimated states. The alternative check uses $z_t = \Phi^{-1}\{\sum_i w_{t|t-1,i}\Phi[(y_t - \mu_i)/\sigma_i]\}$ and refits every bootstrap sample. The model-implied probabilities of at most five losses are 0.0040 for the note and 0.0286 for the improved fit, computed by an exact Markov count recursion at fixed fitted parameters. These are predictive calibration warnings, not exact goodness-of-fit p-values after estimation.

Stability. Removing estimated July leaves mean returns near 1.55% and 5.75%, with durations near 13 months. Returning to the earlier May sample gives 1.57% and 5.83%. Hiding each of four high-return episodes, while preserving calendar transitions, leaves two positive-mean patterns but can materially change their durations. The January–February 2021 deletion gives approximately 6.7 and 2.9 months. Precise state labels and durations should therefore be treated as sample-sensitive estimates.

Connection to prior work and sources

This extends Assignment 3's serial-dependence question on the longer July sample. It supports examining changing return conditions, but does not identify their cause or prove the absence of smoothing. Assignment 3 used 113 observations to May; this note uses 115 to July.

Source: supplied `Fund_X_two_regime_model.docx`, September 2026. The stated `COMBined3b.xlsx` was not found. The project CSV was instead independently reconciled month by month to the saved July 2026 fund factsheet, downloaded August 10, 2026; its exact snapshot and hash are retained.

Method references: Hardy (2001), *North American Actuarial Journal* 5(2), 41–53, bibliographic details checked against the author's university profile; statsmodels Markov switching documentation; scikit-learn Gaussian mixture documentation.